# User Authentication module

### User story - USUA001

As a new user, I want to register an account so that I can access the system features

### Acceptance Criteria ACUA001

- The system should allow the user to enter registration details such as username, email, and password.
- The system should ensure that all required fields are filled before submitting the registration form.
- The system should validate that the email address is in the correct format.
- The system should ensure the password meets the required rules (minimum length and valid characters).
- The system should prevent registration if the username or email already exists in the system.
- The system should display a confirmation message after successful account creation.

### User story - USUA002

As a registered user, I want to login to the system so that I can access my account

### Acceptance Criteria ACUA002

- The system should allow the user to enter their username or email and password.
- The system should validate the entered login credentials with the stored user data.
- The system should grant access to the system when the login credentials are correct.
- The system should display an error message if the username or password is incorrect.
- The system should prevent login if the required fields are empty.
- The system should redirect the user to the main menu or dashboard after successful login.

### User story - USUA003

As a guest, I want to access the system without registering so that I can view available movies

### Acceptance Criteria ACUA003

- The system should allow users to enter the system as a guest without creating an account.
- The system should allow guest users to view the list of available movies.
- The system should allow guest users to view movie details such as title, genre, language, and showtime.

- The system should restrict guest users from booking tickets.
- The system should prompt guest users to register or login if they attempt to book a ticket.

**User story - USUA004**

As a system, I want to validate user details so that only valid users can access the system

**Acceptance Criteria ACUA004**

- The system should verify that the entered username or email exists in the system during login.
- The system should validate that the entered password matches the stored user password.
- The system should check that all required login fields are filled before processing the request.
- The system should display an error message if the user details are invalid.
- The system should allow access to the system only when the user details are verified successfully.

**User story - USUA005**

As a user, I want to login using my Google Sign-In account so that I can access the system quickly

**Acceptance Criteria ACUA005**

- The system should provide an option for users to log in using their Google account.
- The system should redirect the user to the Google authentication page for login.
- The system should verify the user's Google account credentials through Google authentication.
- The system should allow the user to access the system after successful Google authentication.
- The system should display an error message if Google authentication fails.

**User story - USUA006**

As an administrator, I want to log in using my assigned username and password so that I can manage the movie booking system.

**Acceptance Criteria ACUA001**

- The system should allow the administrator to enter their assigned username and password.
- The system should verify the administrator credentials with the stored administrator account details.
- The system should allow access to the administrator dashboard after successful login.
- The system should display an error message if the administrator username or password is incorrect.
- The system should prevent access to administrator features if authentication fails.

**User story - USUA007**

As an officer, I want to log in using my username and password so that I can manage ticket bookings at the theatre counter

**Acceptance Criteria ACUA007**

- The system should allow the officer to enter their username and password.
- The system should verify the officer credentials with the stored officer account details.
- The system should allow the officer to access the officer dashboard after successful login.
- The system should display an error message if the officer username or password is incorrect.
- The system should restrict access to officer functions if authentication fails.

## Payment Processing module

**User story - USPP001**

As a customer, I want to make payment so that I can confirm my ticket booking

**Acceptance Criteria ACPP001**

- The system should allow the customer to choose a payment method to complete the ticket booking.
- The system should display the total payment amount before confirming the payment.
- The system should process the payment after the customer confirms the selected payment method.
- The system should confirm the ticket booking only after the payment is successful.
- The system should display a payment confirmation message after successful payment.

**User story - USPP001**

As a customer, I want to pay using a credit or debit card so that I can complete my booking

**Acceptance Criteria - ACPP002**

- The system should allow the customer to select credit card or debit card as the payment method.
- The system should allow the customer to enter card details such as card number, cardholder name, expiry date, and CVV.
- The system should validate the entered card details before processing the payment.
- The system should process the payment after successful validation of the card details.
- The system should display a confirmation message after the payment is completed successfully.

**User story - USPP003**

As a customer, I want to pay using UPI so that I can make quick payments

**Acceptance Criteria - ACPP003**

- The system should allow the customer to select UPI as the payment method.
- The system should allow the customer to enter their UPI ID for payment.
- The system should validate the entered UPI ID before processing the payment.
- The system should process the payment after successful UPI authentication.
- The system should display a payment confirmation message after the payment is completed successfully.

**User story - USPP004**

As an officer, I want to accept cash payments so that customers can pay at the theatre counter

**Acceptance Criteria - ACPP004**

- The system should allow the officer to select **cash as the payment method** from the console menu.
- The system should display the **total ticket amount** to be paid by the customer.
- The system should allow the officer to **confirm that the cash payment has been received**.
- The system should complete the ticket booking after the cash payment is confirmed.
- The system should display a **payment confirmation message and booking details** after successful payment.

**User story – USPP005**
As a customer, I want to receive payment confirmation so that I know my booking is successful

**Acceptance Criteria – ACPP005**

- The system should display a payment successful message after the payment is completed.
- The system should display the booking details such as movie name, seat number, and showtime after successful payment.
- The system should generate and display a transaction ID for the payment.
- The system should confirm that the ticket booking is completed successfully.
- The system should allow the customer to view the booking confirmation in the console.

**User story – USPP006**
As a customer, I want to see an error message if payment fails so that I can retry payment

**Acceptance Criteria – ACPP006**

- The system should detect when a payment attempt fails during the payment process.
- The system should display a clear error message indicating that the payment has failed.
- The system should prevent ticket booking confirmation when the payment fails.
- The system should allow the customer to retry the payment process.
- The system should return the user to the payment selection menu to attempt payment again.

**User story – USPP007**
As a system, I want to store payment transaction details so that payment records can be tracked

**Acceptance Criteria – ACPP007**

- The system should record the transaction ID for each successful payment.
- The system should store payment details such as movie name, seat number, payment method, and amount paid.
- The system should save the transaction record after successful payment processing.
- The system should allow the stored payment details to be used for booking confirmation and tracking.

- The system should maintain a list of transaction records within the system.

## Ticket cancellation module

### User story – USTC001

As a customer, I want to cancel my booked ticket so that I can free the seat if I cannot attend the show

### Acceptance Criteria – ACTC001

- The system should allow the customer to select the **cancel ticket option** from the console menu.
- The system should prompt the customer to **enter the booking ID or ticket ID**.
- The system should verify that the **booking exists in the system**.
- The system should allow the customer to **confirm the ticket cancellation**.
- The system should cancel the ticket after confirmation.

### User story – USTC002

As a system, I want to update seat availability after ticket cancellation so that the seat can be booked again

### Acceptance Criteria – ACTC002

- The system should identify the **seat number linked to the cancelled ticket**.
- The system should mark the **seat as available after cancellation**.
- The system should update the **seat availability list in the system**.
- The system should allow other users to **book the seat again**.

### User story – USTC003

As an officer, I want to cancel tickets for customers at the theatre counter so that booking errors can be corrected

### Acceptance Criteria – ACTC003

- The system should allow the officer to **select the cancel ticket option from the console menu**.
- The system should allow the officer to **enter the booking ID or customer details**.
- The system should verify the **ticket booking details**.
- The system should allow the officer to **confirm the cancellation request**.
- The system should cancel the ticket and update the booking status.

### User story – USTC004

As a customer, I want to receive confirmation after cancelling my ticket so that I know the cancellation was successful

**Acceptance Criteria – ACTC004**

- The system should display a **ticket cancellation success message** after the ticket is cancelled.
- The system should display the **cancelled ticket details such as movie name, seat number, and showtime**.
- The system should confirm that the **ticket has been successfully cancelled in the system**

**User story – USTC005**

As a customer, I want to receive a refund after cancelling my ticket so that my payment is returned

**Acceptance Criteria – ACTC005**

- The system should check whether the **ticket was paid using an online payment method**.
- The system should calculate the **refund amount for the cancelled ticket**.
- The system should process the **refund to the original payment method**.
- The system should display a **refund confirmation message** after the refund is processed.

**User story – USTC006**

As a system, I want to update the transaction status after ticket cancellation so that refund information is recorded

**Acceptance Criteria – ACTC006**

- The system should identify the **transaction related to the cancelled ticket**.
- The system should update the **transaction status to cancelled or refunded**.
- The system should store the **refund details in the transaction record**.
- The system should maintain the **updated transaction information for tracking purposes**.